

Nonlinear Disturbance Observers — a working introduction

Tutorial note, with full intermediate steps — for the Mecanum ASMC design

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1 What problem are we solving?

Our plant model is imperfect: friction and wheel slip act on the platform but we cannot measure them directly (we'd need contact-patch slip and roller spin, which aren't available on a real robot). The idea of a *disturbance observer* is to lump everything the model gets wrong into one signal d , then **estimate d online using only measurable quantities** (here: the platform velocity v and the commanded torque τ). Once we have an estimate $\hat{d}$, the controller can cancel it — so the robust part of the controller has far less to fight, which means less chattering and tighter tracking.

Goal. Build a dynamical system (the “observer”) whose output $\hat{d}(t)$ converges to the true disturbance $d(t)$, using only signals we already have, and *without* differentiating noisy measurements.

2 Setup and notation

Write the plant in **control-affine** form (state x , input u , known functions f, g):

$$\dot{x} = \underbrace{f(x) + g(x)u}_{\text{known nominal } \Phi(x,u)} + d, \quad (1)$$

where d collects all unmodeled effects (our friction/slip). For the Mecanum platform $x = v = [V_x, V_y, \Omega]^\top$, $u = \tau$ (wheel torques), and d is the lumped friction wrench in acceleration units. Everything that follows holds for general f, g ; we specialise at the end.

3 Why we can't just compute d directly

Rearranging (1) gives $d = \dot{x} - f(x) - g(x)u$. The right-hand side needs $\dot{x}$ — the *time-derivative of the measured state*. Differentiation amplifies noise: a measurement ripple $a \sin(\omega t)$ becomes $a\omega \cos(\omega t)$ after differentiating, so high-frequency sensor noise is multiplied by ω and swamps the estimate. **So we need an estimator that never differentiates x .** That single requirement is what forces the clever structure below.

4 The auxiliary-variable trick — derived step by step

We introduce an internal variable z (the observer's own state) and *define* the estimate as

$$\hat{d} = z + p(x), \quad (2)$$

for some function $p(x)$ we get to choose. Define the **gain** as its Jacobian $L(x) := \frac{\partial p}{\partial x}$. We are free to design the dynamics $\dot{z}$; we will pick them so that the troublesome $\dot{x}$ cancels. Differentiate (2) using the chain rule:

$$\dot{\hat{d}} = \dot{z} + \frac{\partial p}{\partial x} \dot{x} = \dot{z} + L(x) \dot{x}. \quad (3)$$

Substitute the plant (1), $\dot{x} = f + gu + d$:

$$\dot{\hat{d}} = \dot{z} + L(x)[f(x) + g(x)u] + L(x)d. \quad (4)$$

Now **choose** the observer dynamics to cancel the known terms (and the as-yet-unknown d via $\hat{d}$):

$$\boxed{\dot{z} = -L(x)z - L(x)[p(x) + f(x) + g(x)u]} \quad (5)$$

Note $\dot{z}$ uses only x, u and the known f, g, p — **no** $\dot{x}$. Substitute (5) into (4):

$$\begin{aligned} \dot{\hat{d}} &= \left[-Lz - L(p + f + gu) \right] + L(f + gu) + Ld \\ &= -Lz - Lp - \underbrace{L(f + gu) + L(f + gu)}_{=0} + Ld \\ &= -L(z + p) + Ld = -L\hat{d} + Ld = L(d - \hat{d}). \end{aligned} \quad (6)$$

So the estimate obeys $\dot{\hat{d}} = L(d - \hat{d})$. Introduce the **estimation error** $e := d - \hat{d}$. Then $\dot{e} = \dot{d} - \dot{\hat{d}} = \dot{d} - L(x)e$, i.e.

$$\dot{e} = -L(x)e + \dot{d}. \quad \text{(the master equation)} \quad (7)$$

Everything about the observer's behaviour is read from this one linear-in- e equation.

5 Reading the master equation

Equation (7) says the error is the output of a stable filter $\dot{e} = -Le$ driven by the disturbance's rate $\dot{d}$. Three consequences, with L a positive-definite ("stable") gain:

- **Convergence rate** $= \lambda_{\min}(L)$ (how fast the filter forgets). Bigger $L \Rightarrow$ faster.
- **Steady residual** $\approx \|\dot{d}\|/\lambda_{\min}(L)$. The error settles to a ball whose size is the disturbance *rate* divided by the gain.
- **Noise sensitivity** grows with L (because L multiplies the measured x in $\hat{d} = z + Lx$). This is the unavoidable bandwidth-vs-noise trade.

Key takeaway: the error goes to *exactly* zero only when $\dot{d} \rightarrow 0$ (a constant disturbance). For any time-varying d , a constant linear gain leaves a residual $\propto \dot{d}/L$.

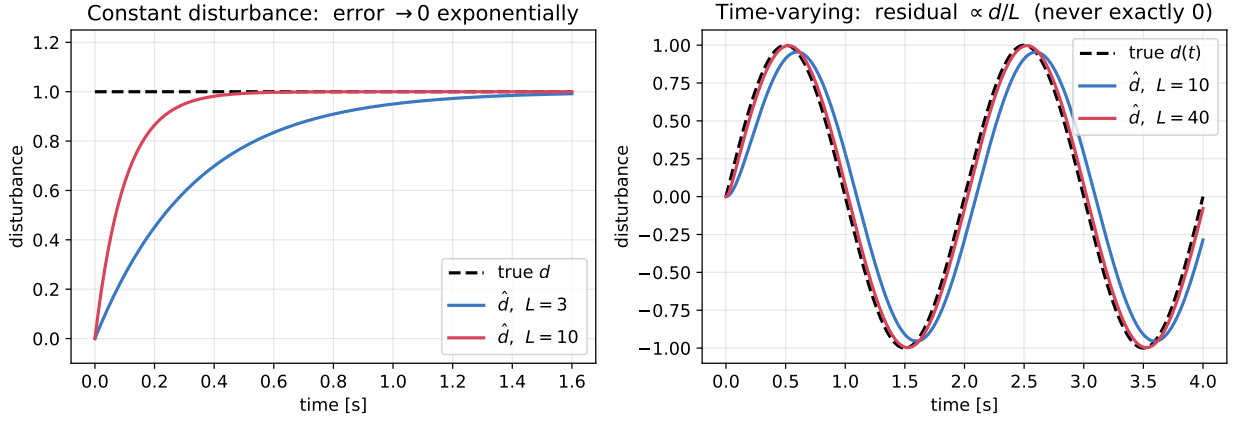
6 A concrete scalar example (with picture)

Take the simplest plant $\dot{x} = u + d$ (so $f = 0$, $g = 1$). With $p(x) = Lx$ the observer (2), (5) becomes

$$\hat{d} = z + Lx, \quad \dot{z} = -L(z + Lx) - Lu,$$

and the master equation is $\dot{e} = -Le + \dot{d}$.

- **Constant** d ($\dot{d} = 0$): $e(t) = e(0)e^{-Lt} \rightarrow 0$, exponentially, faster for larger L .
- **Sinusoid** $d = A \sin \omega t$: the steady error amplitude is $A\omega/\sqrt{L^2 + \omega^2} \approx A\omega/L$ for $L \gg \omega$ — a residual proportional to $\dot{d}$ and shrinking with L , but never zero.



Left: a constant disturbance is estimated with error decaying to zero (faster at higher gain). Right: a time-varying disturbance is tracked with a lag/residual that shrinks as the gain grows but stays nonzero. This picture *is* the master equation (7) in action.

7 Choosing the gain function — the design families

The only freedom is $p(x)$ (equivalently $L(x)$) and, more generally, the *injection function* that replaces $-Le$. Each choice corresponds to an assumption about how d behaves, and buys a different convergence guarantee.

Family	Injection / structure	Error guarantee
Linear constant gain (Chen)	$\dot{e} = -Le + \dot{d}$, L const PD	exponential if $\dot{d} = 0$; residual $\ \hat{d}\ /\lambda_{\min}(L)$ otherwise
Extended State Observer (ADRC)	model $d, \dot{d}, \dots$ as extra integrator states	zero residual for <i>polynomial</i> d up to the modelled order; error $\propto d^{(n)}/\omega_o^n$
Super-twisting sliding-mode DO	$-k_1 e ^{1/2}\text{sign}(e) - k_2\int\text{sign}(e)$	<i>exact, finite-time</i> for $ \dot{d} \leq$ known bound; continuous estimate
Internal-model / ex-system	embed an oscillator at d 's frequency	exact asymptotic for that disturbance <i>class</i> (e.g. periodic)
High-gain	large constant L	approximate ($\propto 1/L$), fast, noise-prone

One intermediate step on the ESO idea. “Modelling d as a state” means: append d to the state and assume $\dot{d} \approx 0$ (order 1). The observer then has one integrator devoted to d , and by (7) the residual is $\propto \dot{d}$. If instead you append *both* d and $\dot{d}$ and assume $\ddot{d} \approx 0$ (order 2), the observer estimates the slope too, and the residual drops to $\propto \ddot{d}$ — one order smaller. Each extra integrator zeros one more derivative of d , at the cost of more phase lag and noise sensitivity. **So you match the observer order to how smooth d actually is.**

8 The internal-model principle (the unifying idea)

To drive the estimation error to *exactly* zero, the observer must contain a model of how the disturbance is generated. Constant $d \rightarrow$ one integrator; ramp/polynomial $d \rightarrow$ several integrators; periodic $d \rightarrow$ an oscillator at that frequency; “only bounded rate” $d \rightarrow$ a sliding-mode (super-twisting) injection.

A plain linear gain assumes nothing beyond “ d is slowly varying,” which is why it leaves a residual. Any *correct* structural assumption you add buys exact convergence for that class; any *wrong* assumption just adds lag and noise.

9 Specialising to the Mecanum platform

State $x = v = [V_x, V_y, \Omega]^\top$, nominal (no-slip) acceleration $\Phi(v, \tau) = M^{-1}(B\tau + C(v))$, and lumped disturbance $\delta = M^{-1}(G_{\text{fric}} - B\tau)$ in acceleration space. Because the allocation $B \in \mathbb{R}^{3 \times 4}$ has full row rank, $BB^+ = I_3$ and any $\delta \in \mathbb{R}^3$ is **matched** (cancellable through the same channel as the control). The observer ($p(v) = Lv$, so L constant) is

$$\hat{d} (= \hat{\delta}) = z + Lv, \quad \dot{z} = -L(\hat{\delta} + \Phi(v, \tau)), \quad (8)$$

and the controller adds a $-\hat{\delta}$ feed-forward to its equivalent control. It uses only v and τ — never roller spin or slip.

Stability proof, with every step

(a) Observer error. From (7) with L constant, $\dot{e} = -Le + \dot{\delta}$. Take the Lyapunov function $V_o = \frac{1}{2}e^\top e$ (a “stored error energy”). Differentiate:

$$\dot{V}_o = e^\top \dot{e} = e^\top (-Le + \dot{\delta}) = -e^\top Le + e^\top \dot{\delta} \leq -\lambda_{\min}(L)\|e\|^2 + \|e\| \|\dot{\delta}\|.$$

The right side is negative whenever $\|e\| > \|\dot{\delta}\|/\lambda_{\min}(L)$, so V_o shrinks until $\|e\| \leq \|\dot{\delta}\|/\lambda_{\min}(L)$ (input-to-state stable; exponential to 0 if $\dot{\delta} \rightarrow 0$).

(b) Closed loop. Let the sliding surface be $s = (v - v_{des}) + \Lambda e_p$ (e_p = position error). With the control choosing the nominal acceleration $a_{des} = \dot{v}_{des} - \Lambda(v - v_{des}) - \hat{\delta} - K \tanh(s/\varepsilon) - \Lambda_s s$ and $BB^+ = I_3$, one finds (substituting $\dot{v} = \Phi + \delta$ and $\Phi = a_{des}$):

$$\dot{s} = \dot{v} - \dot{v}_{des} + \Lambda(\dots) = \delta - \hat{\delta} - K \tanh(s/\varepsilon) - \Lambda_s s = e - K \tanh(s/\varepsilon) - \Lambda_s s.$$

Take $V_s = \frac{1}{2}s^\top s$. Outside the boundary layer $s^\top \tanh(s/\varepsilon) \geq \|s\|$, so

$$\dot{V}_s = s^\top \dot{s} - K s^\top \tanh(s/\varepsilon) - \Lambda_s \|s\|^2 \leq \|s\|(\|e\| - K) - \Lambda_s \|s\|^2.$$

Choosing

$$K > \frac{\|\dot{\delta}\|}{\lambda_{\min}(L)} + \eta \quad (\eta > 0)$$

gives $\dot{V}_s \leq -\eta\|s\| - \Lambda_s\|s\|^2 < 0$: s reaches a small boundary layer in finite time and the tracking error follows. **The payoff:** without the observer you’d need $K > \|\delta\|$ (the full disturbance); with it you need only $K > \|\dot{\delta}\|/\lambda_{\min}(L)$ (its rate) — a large reduction in switching gain, hence far less chatter, and the deterministic part of δ is actively cancelled.

10 Which form to pick for the friction disturbance

Our δ has (i) a slowly-varying dominant part (Stribeck + spin coupling), (ii) a periodic component (the sineline forcing repeats at $2\pi/T_{\text{period}}$), and (iii) fast near-discontinuous spikes at zero-slip crossings.

- **Baseline:** linear constant-gain NDOB (the proof above). Simple, robust, provably ISS.
- **If the slow drift residual matters:** go to an order-2 (ESO-style) observer — one extra integrator zeros the ramp term.
- **If the periodic yaw error dominates:** add an internal model (oscillator) at $2\pi/T_{\text{period}}$ to cancel it exactly.
- **The spikes:** no smooth finite-gain observer (and not even super-twisting, whose premise is bounded $\dot{\delta}$) tracks a true discontinuity — raising L there only injects noise. Accept a residual at crossings and let the SMC boundary layer absorb it.

11 Mini-glossary

ISS (input-to-state stable)

the error stays in a bounded ball whose size is set by the “input” here $\dot{d}$; if the input vanishes, the error goes to zero.

Exponential convergence

error decays like $e^{-\lambda t}$.

Finite-time convergence

error becomes *exactly* zero after a finite time (sliding-mode property).

Matched disturbance

one that enters through the same channel as the control, so it can be fully cancelled.

Boundary layer ε

the band around the sliding surface where sign is replaced by $\tanh(\cdot/\varepsilon)$ to avoid chatter; rejection is only approximate inside it.

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